



HUMAN-COMPUTER INTERACTION

THIRD
EDITION

DIX
FINLAY
ABOWD
BEALE

HUMAN COMPUTER INTERACTION

the computer

The Computer

a computer system is made up of various elements

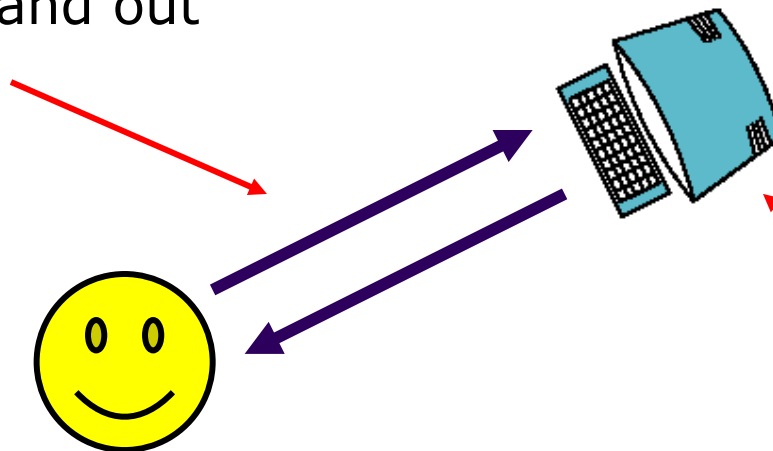
each of these elements affects the interaction

- input devices – text entry and pointing
- output devices – screen (small&large), digital paper
- virtual reality – special interaction and display devices
- physical interaction – e.g. sound, haptic, bio-sensing
- processing – speed of processing, networks

Interacting with computers

to understand human-*computer* interaction
... need to understand computers!

what goes in and out
devices, paper,
sensors, etc.

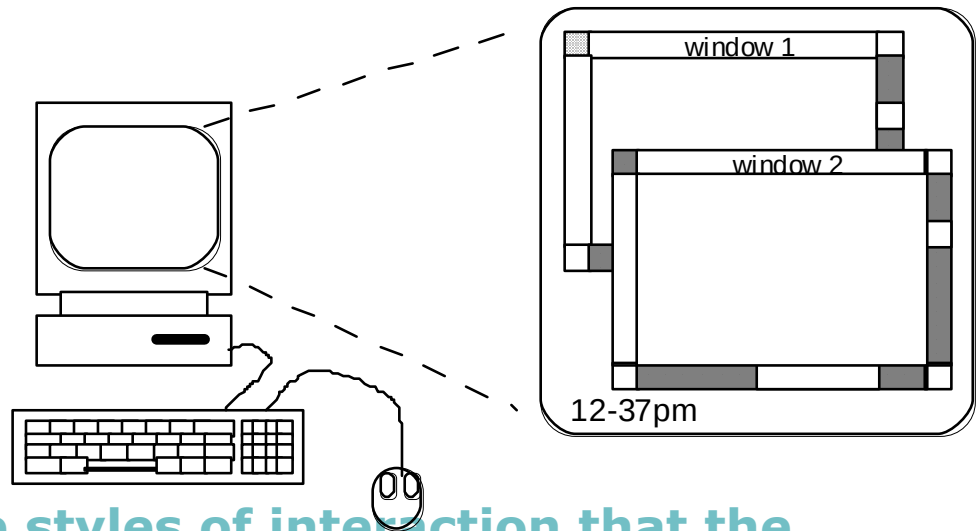


what can it do?
memory, processing,
networks

A 'typical' computer system

?

- screen, or monitor, on which there are windows
- keyboard
- mouse/trackpad
- variations
 - desktop
 - laptop
 - PDA



the devices dictate the styles of interaction that the system supports

If we use different devices, then the interface will support a different style of interaction



How many computers ...

in your house?

- PC
- TV, VCR, DVD, HiFi, cable/satellite TV
- microwave, cooker, washing machine
- central heating
- security system

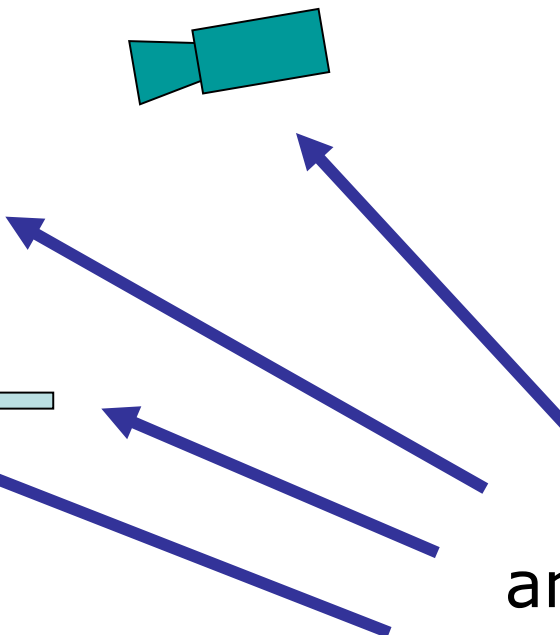
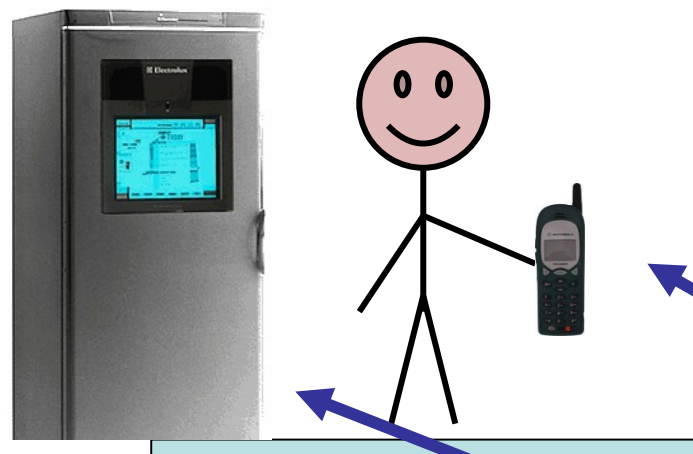
can you think of more?

in your pockets?

- PDA
- phone, camera
- smart card, card with magnetic strip?
- electronic car key
- USB memory

try your pockets and bags

Richer interaction



sensors
and devices
everywhere

text entry devices

keyboards (QWERTY et al.)
chord keyboards, phone pads
handwriting, speech

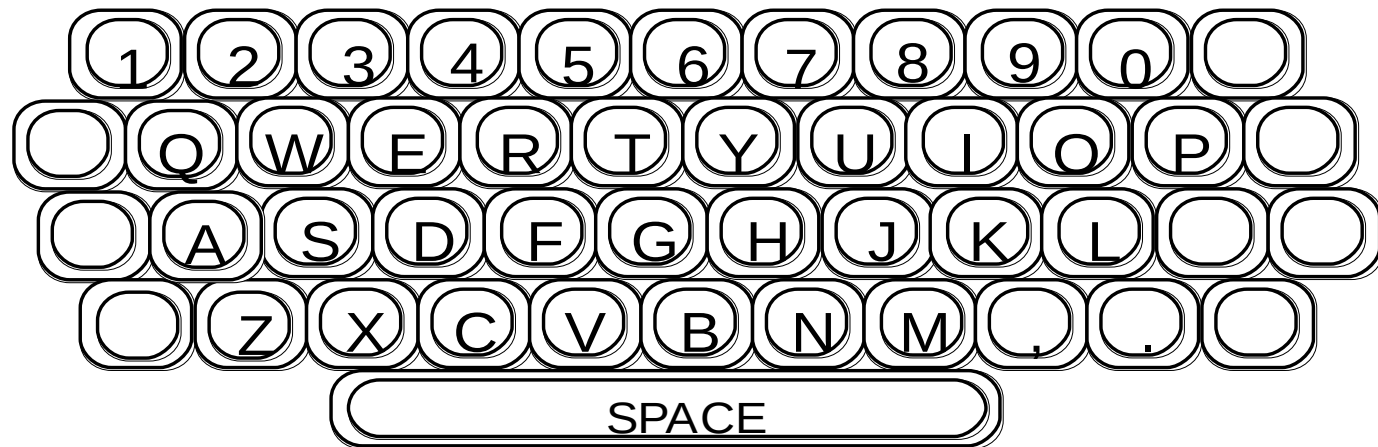
Keyboards

- Most common text input device
- Allows rapid entry of text by experienced users
- Keypress closes connection, causing a character code to be sent
- Usually connected by cable, but can be wireless

layout - QWERTY

- Standardised layout
but ...
 - non-alphanumeric keys are placed differently
 - accented symbols needed for different scripts
 - minor differences between UK and USA keyboards
- QWERTY arrangement not optimal for typing
 - layout to prevent typewriters jamming!
- Alternative designs allow faster typing but large social base of QWERTY typists produces reluctance to change.

QWERTY (ctd)



alternative keyboard layouts

Alphabetic

- keys arranged in alphabetic order
- not faster for trained typists
- not faster for beginners either!

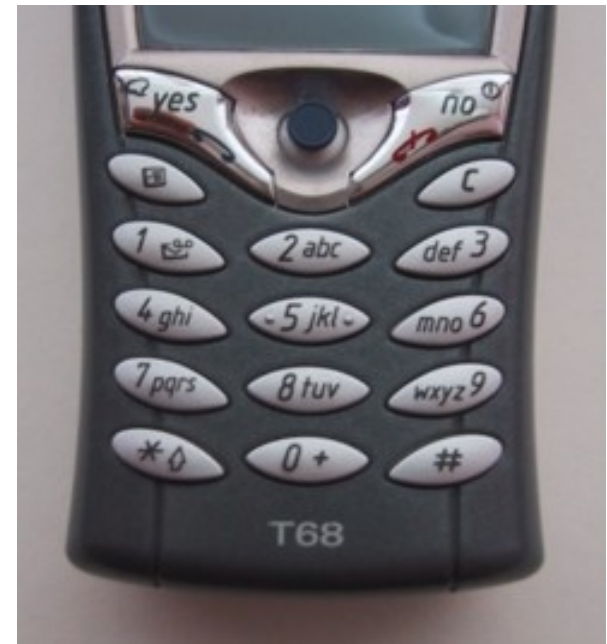
Dvorak

- common letters under dominant fingers
- biased towards right hand
- common combinations of letters alternate between hands
- 10-15% improvement in speed and reduction in fatigue
- But - large social base of QWERTY typists produce market pressures not to change

phone pad and T9 entry

- use numeric keys with multiple presses
 - 2 - a b c
 - 3 - d e f
 - 4 - g h i
 - 5 - j k l
 - 6 - m n o
 - 7 - p q r s
 - 8 - t u v
 - 9 - w x y z

hello = 4433555[pause]555666
surprisingly fast!
- T9 predictive entry
 - type as if single key for each letter
 - use dictionary to 'guess' the right word
 - hello = 43556 ...
 - but 26 -> menu 'am' or 'an'



Handwriting recognition

- Text can be input into the computer, using a pen and a digitizing tablet
 - natural interaction
- Technical problems:
 - capturing all useful information - stroke path, pressure, etc. in a natural manner
 - segmenting joined up writing into individual letters
 - interpreting individual letters
 - coping with different styles of handwriting
- Used in PDAs, and tablet computers ...
... *leave the keyboard on the desk!*

Speech recognition

- Improving rapidly
- Most successful when:
 - single user – initial training and learns peculiarities
 - limited vocabulary systems
- Problems with
 - external noise interfering
 - imprecision of pronunciation
 - large vocabularies
 - different speakers

positioning, pointing and drawing

mouse, touchpad
trackballs, joysticks etc.
touch screens, tablets
eyegaze, cursors

the mouse (ctd)

Mouse located on desktop

- requires physical space
- no arm fatigue

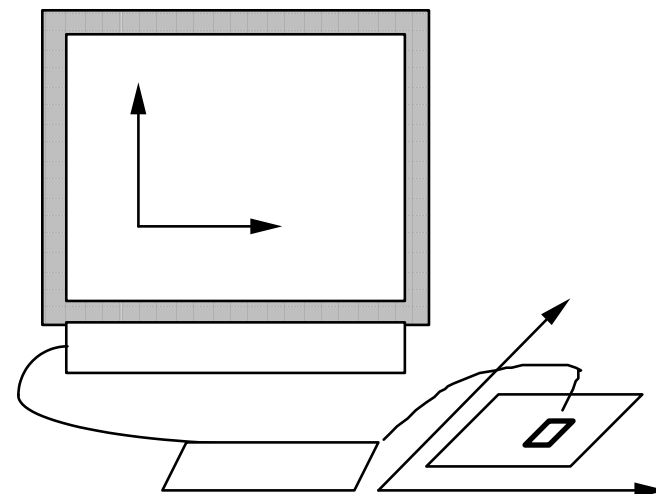
Relative movement only is detectable.

Movement of mouse moves screen cursor

Screen cursor oriented in (x, y) plane,
mouse movement in (x, z) plane ...

... an *indirect* manipulation device.

- device itself doesn't obscure screen, is accurate and fast.
- hand-eye coordination problems for novice users



Touch-sensitive screen

- Detect the presence of finger or stylus on the screen.
 - works by interrupting matrix of light beams, capacitance changes or ultrasonic reflections
 - *direct* pointing device
- Advantages:
 - fast, and requires no specialised pointer
 - good for menu selection
 - suitable for use in hostile environment: clean and safe from damage.
- Disadvantages:
 - finger can mark screen
 - imprecise (finger is a fairly blunt instrument!)
 - difficult to select small regions or perform accurate drawing
 - lifting arm can be tiring

Stylus and light pen

Stylus

- small pen-like pointer to draw directly on screen
- may use touch sensitive surface or magnetic detection
- used in PDA, tablets PCs and drawing tables

Light Pen

- now rarely used
- uses light from screen to detect location

BOTH ...

- very direct and obvious to use
- but can obscure screen

display devices

bitmap screens (CRT & LCD)
large & situated displays
digital paper



Health hints ...

- do not sit too close to the screen
 - do not use very small fonts
 - do not look at the screen for long periods without a break
 - do not place the screen directly in front of a bright window
 - work in well-lit surroundings
- ★ Take extra care if pregnant.
but also posture, ergonomics, stress

situated displays

- displays in 'public' places
 - large or small
 - very public or for small group
- display only
 - for information relevant to location
- or interactive
 - use stylus, touch sensitive screen
- in all cases ... the location matters
 - meaning of information or interaction is related to the location

virtual reality and 3D interaction

positioning in 3D space
moving and grasping
seeing 3D (helmets and caves)

positioning in 3D space

- data glove
 - fibre optics used to detect finger position
- VR helmets
 - detect head motion and possibly eye gaze
- whole body tracking
 - accelerometers strapped to limbs or reflective dots and video processing

physical controls, sensors etc.

special displays and gauges

sound, touch, feel, smell

physical controls

environmental and bio-sensing

Sounds

- beeps, bongs, clonks, whistles and whirrs
- used for error indications
- confirmation of actions e.g. keyclick

Touch, feel, smell

- touch and feeling important
 - in games ... vibration, force feedback
 - in simulation ... feel of surgical instruments
 - called *haptic* devices (relating to the sense of touch, in particular relating to the perception and manipulation of objects using the senses of touch).
- texture, smell, taste
 - current technology very limited

Environment and bio-sensing

- sensors all around us
 - car courtesy light – small switch on door
 - ultrasound detectors – security, washbasins
 - RFID security tags in shops
 - temperature, weight, location
- ... and even our own bodies ...
 - iris scanners, body temperature, heart rate, galvanic skin response, blink rate